

README

Helix OTA — Enterprise-Grade Universal OTA Update System

Field	Value
Repository	HelixDevelopment/helix_ota
License	Apache 2.0
Language	Go 1.22+ (Server/SDK), Kotlin (Android Client), TypeScript/React (Dashboard)
Status	Planning & Research Phase

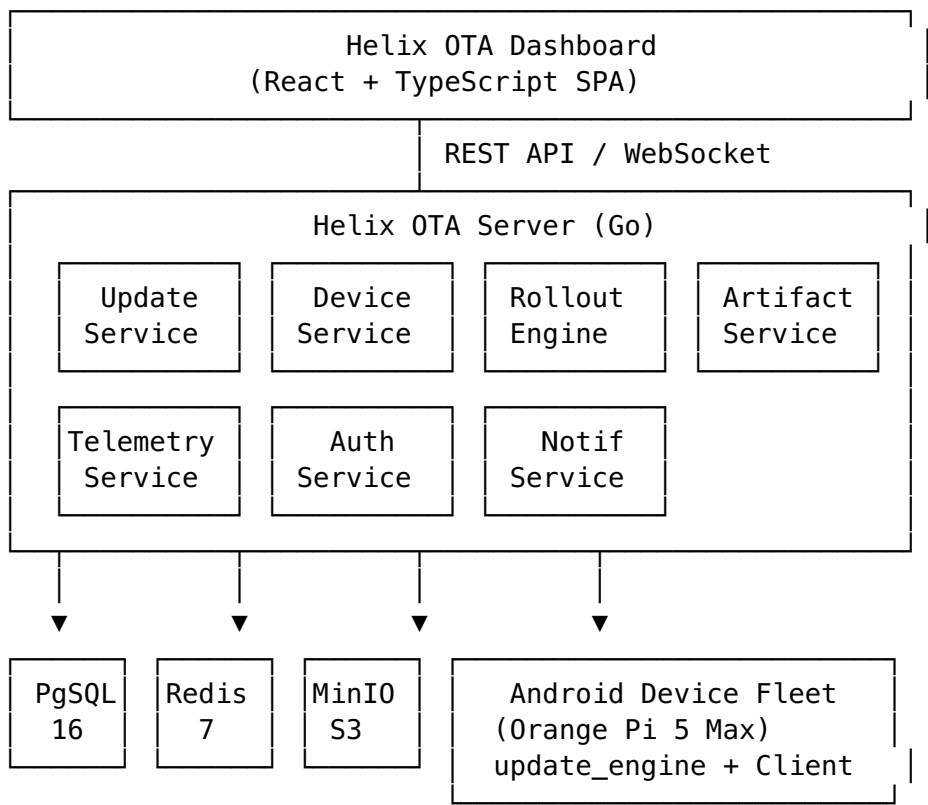
Overview

Helix OTA is a universal, enterprise-grade Over-The-Air update system designed to deliver safe, reliable, and verifiable OTA updates to any operating system. Built on the principles of **universality, generic design, maximum decoupling, and extensibility**, it starts with full Android 15 support (targeting Orange Pi 5 Max / RK3588) and expands to Linux, Windows, and beyond.

Key Features

- **Safe OTA Updates** — SHA-256 + RSA-4096 artifact verification, A/B partition protection, automatic rollback
- **Phased Rollouts** — Deploy updates to 5%, 10%, 30%, 50%, 100% of fleet with auto-rollback on failure
- **Universal Architecture** — Plugin-based OS adapter system supporting Android, Linux, Windows, and more
- **Enterprise Dashboard** — Upload artifacts, manage rollouts, monitor fleet health, analyze telemetry
- **Full Containerization** — All infrastructure containerized via vasic-digital/containers submodule
- **Constitution Compliance** — Follows HelixConstitution anti-bluff, mutation testing, nano-detail documentation mandates

Architecture



Version Roadmap

Version	Name	Focus	Status
1.0.0	MVP	Android 15 OTA (RK3588), Server, Dashboard, Rollouts	 Planning
1.0.1	Rollback	Multi-version rollback, server-triggered rollback	 Planned
1.0.2	Delta Updates	Incremental OTA, 60-90% bandwidth reduction	 Planned
1.1.0	Linux Support	Ubuntu/Debian, Fedora/rpm-ostree, Arch, Generic A/B	 Planned
1.2.0	Windows Support	Windows Service client, MSI/MSIX distribution	 Planned

Version	Name	Focus	Status
2.0.0	Multi-OS Universal	Plugin architecture, all OS types, adapter SDK	 Planned

Documentation Index

1.0.0 MVP Documentation

Document	Path	Description
Version Roadmap	docs/VERSION_ROADMAP.md	Complete version roadmap
System Architecture	docs/architecture/SYSTEM_ARCHITECTURE.md	Full system architecture design
REST API Specification	docs/api/REST_API_SPECIFICATION.md	All API endpoints
Database Schema	docs/database/DATABASE_SCHEMA.md	PostgreSQL schema DDL
Security Architecture	docs/security/SECURITY_ARCHITECTURE.md	Security design & threat model
Testing Strategy	docs/testing/TESTING_STRATEGY.md	Complete testing approach
Deployment Guide	docs/deployment/DEPLOYMENT_GUIDE.md	Docker/K8s deployment
OTA Systems Research	docs/research/OTA_SYSTEMS_RESEARCH.md	Existing OTA solutions survey
Submodule Specs	docs/submodules/NEW_SUBMODULE_SPECIFICATIONS.md	New submodule definitions
Server Implementation	server/SERVER_IMPLEMENTATION.md	Go server design
Android Client Design	client/android/ANDROID_CLIENT_DESIGN.md	Android 15 client

Diagrams

Diagram	Path
High-Level Architecture	

Diagram	Path
	docs/diagrams/HIGH_LEVEL_ARCHITECTURE.md
Update Flow	docs/diagrams/UPDATE_FLOW.md
Rollout Flow	docs/diagrams/ROLLOUT_FLOW.md
Client State Machine	docs/diagrams/CLIENT_STATE_MACHINE.md
Artifact Validation	docs/diagrams/ARTIFACT_VALIDATION.md
Database ER	docs/diagrams/DATABASE_ER.md
Security Layers	docs/diagrams/SECURITY_LAYERS.md
Container Architecture	docs/diagrams/CONTAINER_ARCHITECTURE.md

Future Version Documentation

Version	Document	Path
1.0.1	Rollback Design	1.0.1-rollback/docs/ROLLBACK_DESIGN.md
1.0.2	Delta Updates Design	1.0.2-delta-updates/docs/DELTA_UPDATES_DESIGN.md
1.1.0	Linux OTA Research	1.1.0-linux-support/docs/LINUX_OTA_RESEARCH.md
1.2.0	Windows OTA Research	1.2.0-windows-support/docs/WINDOWS_OTA_RESEARCH.md
2.0.0	Multi-OS Universal Design	2.0.0-multi-os-universal/docs/MULTI_OS_UNIVERSAL_DESIGN.md

Submodules

Existing vasic-digital Modules Used

Module	Purpose
auth	JWT authentication & RBAC
database	PostgreSQL adapter, migrations, repository pattern
cache	Two-level caching (memory + Redis)
observability	Distributed tracing, metrics, structured logging
security	TLS configuration, certificate management
middleware	HTTP middleware chain

Module	Purpose
config	Configuration management
EventBus	Internal event publishing
Storage	S3-compatible artifact storage
ratelimiter	API rate limiting
concurrency	Worker pools, semaphores
recovery	Circuit breakers, fault tolerance
containers	Container orchestration substrate

New HelixDevelopment Submodules

Submodule	Purpose	Version
helix-ota-server	OTA management server	1.0.0
helix-ota-client-sdk	Platform-agnostic Go client SDK	1.0.0
helix-ota-android	Android system app	1.0.0
helix-ota-dashboard	React management dashboard	1.0.0
helix-update-engine	Universal update execution engine	1.0.0
helix-artifact-validator	Artifact validation library	1.0.0
helix-rollout-engine	Phased rollout decision engine	1.0.0
helix-device-identity	Device identity & certificate management	1.0.0

Constitution Compliance

This project follows the [HelixConstitution](#) mandates:

- §1 Test coverage mandatory for every change
- §1.1 Mutation testing (85% mutation score target)
- §7.1 NO BLUFF — positive-evidence-only validation
- §11.4.108 Four-layer fix verification (SOURCE → ARTIFACT → RUNTIME → USER-VISIBLE)
- §6 Documentation up to nano-details

License

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Organizations

- HelixDevelopment — Product/application layer
- vasic-digital — Foundational modules & infrastructure